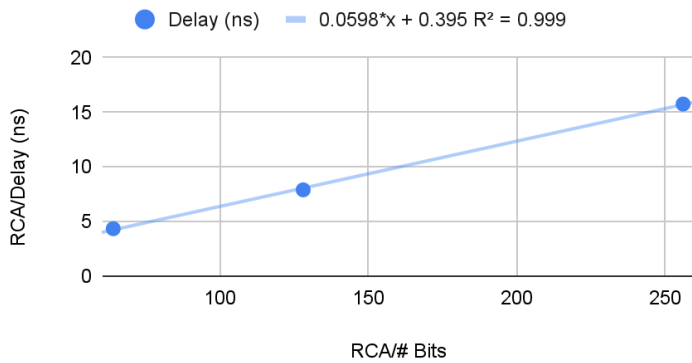
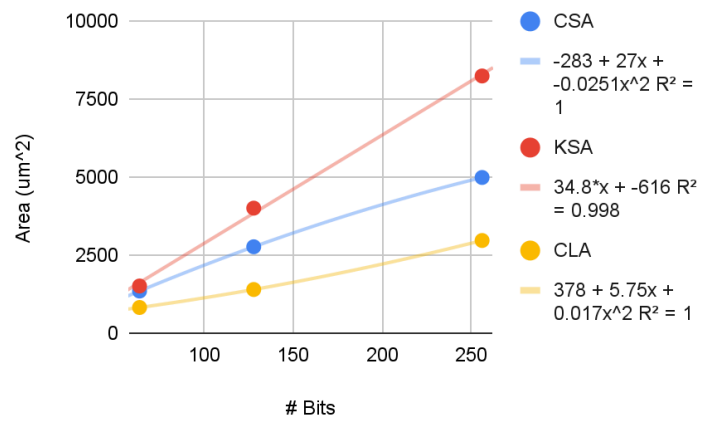
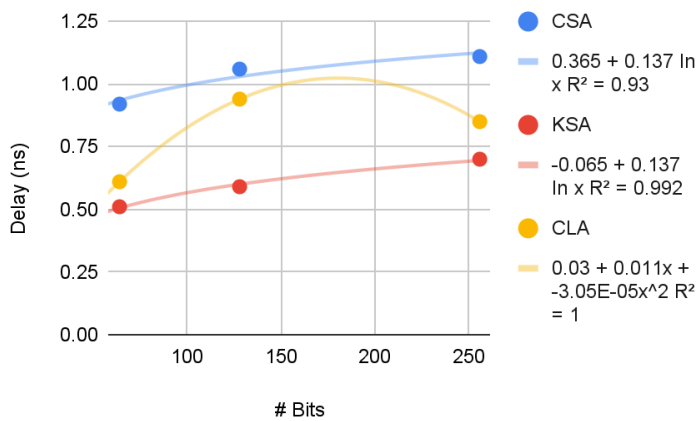
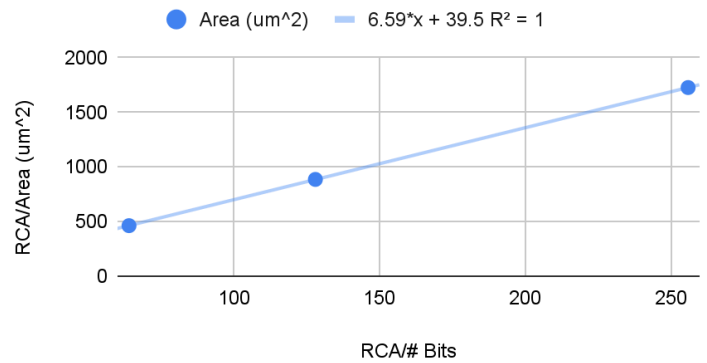


Bits Vs Delay



Bits Vs Area



Delay Analysis

I used the trend line function in Google Sheets, and changed the value of the trend line by the R^2 value.

1. RCA: The trend line of the RCA was $0.0598x + 0.395$, and the R^2 value is 0.999, which indicates that the trend line is reliable. Thus, the delay of RCA is $O(n)$
2. CSA: The trend line of the CSA was $0.365 + 0.137 \ln(x)$, and the R^2 value is 0.93, which indicates that the trend line is reliable. Thus, the delay of CSA is $O(\ln(n))$
3. KSA: The trend line of the KSA was $-0.065 + 0.137 \ln(x)$, and the R^2 value is 0.992, which indicates that the trend line is reliable. Thus, the delay of KSA is $O(\ln(n))$
4. CLA: The trend line of the CLA was $0.03 + 0.011x - 3.05E-05x^2$, and the R^2 value is 1, which indicates that the trend line is reliable. Thus, the delay of CLA is $O(n^2)$. However, I believe the delay of 128 bits might have an error. Even though I decreased the time constraints up to 0.01 ns, the delay did not go down below 0.9. If the 128-bit delay decreased or the 256-bit delay increased, the $O(\ln(n))$ notation would be fitted.

Final Timing Constraints

1. RCA

The original data from the TCL file required a 4ns delay for a 64-bit RCA. The first timing report shows an arrival time of 4.67 ns, which is a -0.67 ns violation. I reduced the required time to 3 ns to check whether the arrival time cannot exceed 4 ns. Turns out, 4.33 was the minimum arrival time. Thus, I increased the data required time to 4.33 ns, but the arrival time increased to 4.67 ns again. I tested with 3 ns, and the final arrival time was 4.33 ns. The area of this RCA_64 is 460.712 μm^2 . I followed the same process for 128- and 256-bit RCA adders, and I got 7.89 ns and 884.716 μm^2 for 128 bits and 15.76 ns and 1727.138 μm^2 for 256 bits.

- a. RCA 64: Constraint: 3 ns Actual Delay: 4.33 ns
- b. RCA 128: Constraint: 5 ns Actual Delay: 7.89 ns
- c. RCA 256: Constraint: 8 ns Actual Delay: 15.76 ns

2. CSA

- a. CSA 64: Constraint: 0.5 ns Actual Delay: 0.92 ns Area: 1343.566 μm^2
- b. CSA 128: Constraint: 0.5 ns Actual Delay: 1.06 ns Area: 2764.462 μm^2
- c. CSA 256: Constraint: 0.7 ns Actual Delay: 1.11 ns Area 4989.096 μm^2

3. KSA

- a. KSA 64: Constraint: 0.2 ns Actual Delay: 0.51 ns Area: 1505.294 μm^2
- b. KSA 128: Constraint: 0.3 ns Actual Delay: 0.59 ns Area: 4004.63 μm^2
- c. KSA 256: Constraint: 0.3 ns Actual Delay: 0.7 ns Area: 8247.33 μm^2

4. CLA

- a. CLA 64: Constraint: 0.3 ns Actual Delay: 0.61 ns Area: 815.556 μm^2
- b. CLA 128: Constraint: 0.1 ns Actual Delay: 0.94 ns Area: 1392.776 μm^2
- c. CLA 256: Constraint: 0.4 ns Actual Delay: 0.85 ns Area: 2966.166 μm^2